

Supplementary Information for “Torddis: A Real-Time IoT and AI-Based System to Support At-Home Learning and Mitigate Parental Absenteeism”

Miguel J. Hornos^{1*}

^{1*}See main manuscript for full author list and affiliations.

Corresponding author(s). E-mail(s): mhornos@ugr.es;

Abstract

This document contains supplementary information for the manuscript entitled “Torddis: A Real-Time IoT and AI-Based System to Support At-Home Learning and Mitigate Parental Absenteeism”. It provides the detailed acceptance test cases, the review of AI methods surveyed for the distraction-monitoring modules, and the preventive-maintenance plan for the Torddis system prototype. The main manuscript is self-contained; this material is provided for readers interested in reproducing the study.

Keywords: Supplementary Information, Torddis, acceptance testing, AI methods, IoT maintenance

1 Test Cases Used in Acceptance Testing

This supplementary section presents the acceptance test cases that were designed and executed during the Torddis system development, following ISO/IEC/IEEE 12207 and TDDM4IoTS practices.

2 Review of AI Methods for Distraction Monitoring

This supplementary section reviews the AI methods surveyed during the preliminary analysis of the Torddis project. Tables list facial-recognition, facial-expression, drowsiness-detection and object-recognition methods evaluated in the literature.

2.0.1 AI Methods for Monitoring Children’s Distractions

In this stage, the most suitable AI methods were selected to implement the Torddis proposal. These include methods for:

- Facial recognition,
- Facial expression recognition,
- Drowsiness detection, and
- Object recognition.

Facial Recognition Methods.

Facial recognition methods focus on face detection by identifying patterns such as eyes, lips, mouth, and nose, among other facial features. Table 1 lists several methods used for face detection and recognition.

Table 1 Facial recognition methods

Method	Implementation Environment	Purpose	Accuracy
HCC ^a	OpenCV	Face detection in low-light environments (Le & Mohd, 2022).	81.00%
Amazon Rekognition	(not specified)	Gender classification (Goel & Agarwal, 2012).	98.75%
	AWS ^b	Authentication through facial recognition (Girmay, Samsom, & Khattak, 2021).	100%
YOLO ^c , MTCNN ^d , and FaceNet with SVC ^e	Google Colab	Attendance registration based on deep learning (Darapaneni et al., 2020).	99.00%
LBPH ^f	(not specified)	Facial detection from captured images (Garcia, Lacayanga, & Cruz, 2021).	91.72%

^aHaar Cascade Classifier: A fast object detection algorithm that uses Haar-like features to detect faces in images. ^bAmazon Web Services. ^cYou Only Look Once: A real-time object detection algorithm. ^dMulti-task Cascaded Convolutional Neural Network: A face detection algorithm.

^eSupport Vector Classifier: A supervised learning technique commonly used for data classification.

^fLocal Binary Patterns Histogram: A facial recognition algorithm that uses local binary patterns and compares histograms for face identification.

The combination of YOLO with MTCNN and FaceNet with SVC was deemed unsuitable for the proposed system due to its computational complexity and high resource requirements. Additionally, Amazon Rekognition was ruled out as it is a commercial service with a significant cost. Therefore, the selected approach consisted of the *HCC* method along with the *LBPH* algorithm from the *OpenCV* library, as it enables easy customization and offers relatively short recognition times compared to other alternatives.

Facial Expression Recognition Methods.

Facial expression recognition requires an input image from which features are extracted and subsequently compared against a computational model. Table 2 lists several methods used for facial expression detection.

As indicated in Table 2, the CNN (Convolutional Neural Networks) method was selected for the development of Torddis due to its accuracy. However, it is essential to ensure proper training of the model. When selecting the most suitable method or algorithm for implementation in a system, it is important to consider that the accuracy of certain algorithms depends on the dataset used.

Table 2 Facial expression recognition methods

Method	Dataset Used	Tools and Frameworks	Purpose	Accuracy
HCC	Custom	(not specified)	Recognizes seven expressions: happy, sad, angry, scared, disgusted, surprised, and neutral (Lalitha, Aishwarya, Shivakumar, Srilekha, & Kartheek, 2021).	78.00%
CNN	FERC ^a 2013	Python, Keras, TensorFlow, and OpenCV	Recognition of basic human emotions (anger, fear, neutral, happiness, sadness, surprise, etc.) (Kedari, Kapile, Kadole, & Jaikar, 2021).	60.00%
		not specified	Emotion recognition through facial expressions (Jaiswal, Krishnama Raju, & Deb, 2020).	70.14%
		(not specified)	FER ^b -based classification using static images (Singh & Nasoz, 2020).	61.70%
	JAFFE ^c	(not specified)	Detecting emotions through facial expressions (Jaiswal et al., 2020).	97.97%
	CK+ ^d	Python, Keras, TensorFlow, and OpenCV	Recognition of basic human emotions such as: anger, fear, neutral, happiness, sadness, and surprise (Kedari et al., 2021).	99.10%
	CK+	Python, 2-layer Sequential CNN, Leaky ReLU, Softmax, Dropout	Facial emotion detection and generation of motivational quotes for each emotion (Bikku et al., 2022).	85.70%
	(not specified)	CNN with 80 epochs ^e	Automatically generates and categorizes questions, evaluates answers, and tracks performance by providing motivational quotes when detecting student emotions (Silva, Sudasinghe, Hansika, Gamage, & Gamage, 2021).	99.00%
MobileNetV2	(not specified)	(not specified)	Detects a person's emotion through facial expression using an artificial neural network (Kumar & Srivastava, 2021).	89.91%
	(not specified)	Python and TensorFlow	Online evaluation of students' ability to train and deploy deep learning solutions (Ilić et al., 2021).	60.00%

^aFacial Expression Recognition Challenge. ^bFacial Expression Recognition. ^cJapanese Female Facial Expression dataset. ^dCK+ refers to the Extended Cohn-Kanade Dataset. ^eAn epoch refers to one complete pass of the model through the entire training dataset, during which each sample is used once to update the model weights.

Facial Expression Recognition Procedure.

The procedure for facial expression recognition is schematized in Figure 1. The process begins with the capture of an image containing the user's face. Subsequently, facial

features are extracted using computer vision techniques, which are then compared with a previously registered set of images to identify the user.

The computational model employed was trained on a dataset representing the seven universal facial expressions: anger, disgust, fear, happiness, sadness, surprise, and neutrality, thus enabling accurate emotion classification based on visual patterns (Zhang & Yu, 2022).

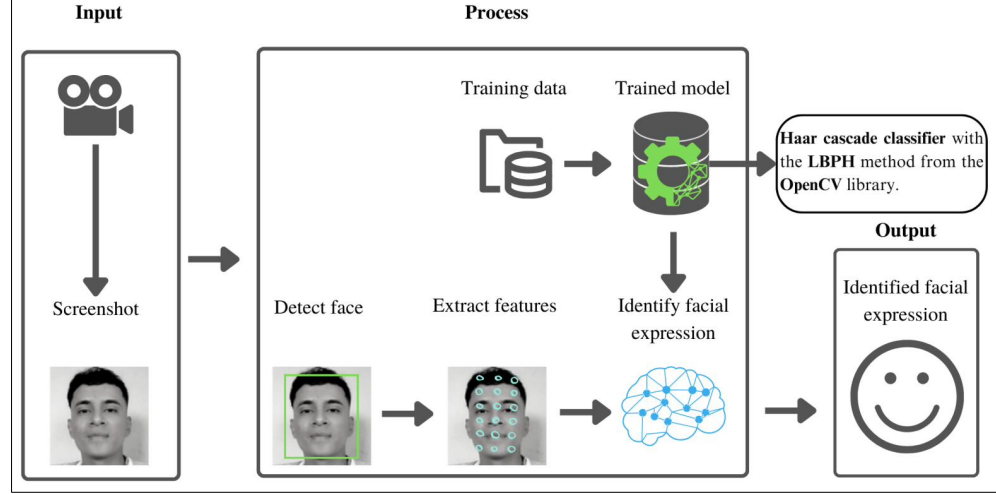


Fig. 1 Facial expression recognition procedure used in the Torddis system.

Drowsiness Detection Methods.

Drowsiness detection methods require an input image containing a human face. These methods perform image preprocessing to extract characteristic eye landmarks and then analyze whether the subject’s eyes are closed. Table 3 summarizes the methods used for drowsiness detection.

Table 3 Drowsiness detection methods.

Algorithm	Purpose	Accuracy
MediaPipe	Face	
Mesh	Detection of facial landmarks with 468 3D points (Shanmugam, Badruddin, & Asirvadam, 2022).	99.87%
SVM	Eye tracking (Altameem, Kumar, Poonia, Kumar, & Saudagar, 2021).	83.25%
Viola-Jones	Drowsiness detection (Teja & Kumar, 2021).	92.00%
CNN	Trained on four gestures: eyes open, eyes closed, yawning, and not yawning (I, Ramli, Azri, Aliff, & Mohammad, 2022).	80.00%

The SVM (Altameem et al., 2021), CNN (I et al., 2022), and Viola-Jones (Teja & Kumar, 2021) algorithms were discarded due to their limitations in accuracy for drowsiness detection and their complexity in real-time implementation. Specifically, SVM and Viola-Jones require intensive preprocessing and struggle to handle variations in lighting and posture, whereas CNNs, although accurate, demand high computational capacity for inference on low-power devices.

For these reasons, the selected algorithm was MediaPipe Face Mesh (Shanmugam et al., 2022), as it offers an optimal balance between accuracy and computational efficiency, enabling real-time drowsiness detection with low latency and without the need for specialized hardware.

Object Recognition Methods

Computer vision enables the automatic detection of the structure and properties of a potential dynamic three-dimensional world. First, an input image containing one or more objects is required. Then, image preprocessing is performed to extract features from the segmented image. Table 4 describes a list of object recognition algorithms.

Based on the information presented in Table 4, the selected algorithm was *Sequential* from the *TensorFlow* library in combination with *Keras* due to its accuracy, optimal performance, ease of training, and widespread use for real-time object recognition. MobileNetV2 combined with SSD was discarded due to its lower object recognition accuracy compared to other models. YOLO was also discarded because of the high GPU (Graphics Processing Unit) requirements for proper operation.

Object Recognition Procedure

This method consists of the following steps:

1. A photograph is taken to capture the objects on the desk or in the area where the child is working.
2. The image is preprocessed and segmented into parts corresponding to potential objects.
3. Feature extraction is performed on the segmented image.
4. Finally, the extracted features are compared with a computational model previously trained on a dataset containing the objects to be recognized.

Computer vision enables the automatic detection of the structure and properties of a potential dynamic three-dimensional world based on one or more two-dimensional images of the scene (Cruz et al., 2013). Figure 2 illustrates the object recognition procedure used in the Torddis system implementation.

The analysis highlights that Torddis outperforms existing market solutions due to its integration of facial and object recognition technologies to effectively monitor distractions. This advantage is reflected in the high acceptance rates among tutors and the positive results observed in students, positioning the system as a superior solution for monitoring and supporting academic tasks at home (Deng, Jiang, Yu, Lu, & Liu, 2024). Following the elicitation of system requirements, the architecture of the Torddis system was initially designed (see Figure 3). Subsequently, with the

Table 4 Algorithms for object recognition.

Algorithm	Technology and Implementation	Purpose	Accuracy
CNN	Python, TensorFlow, Keras, CIFAR-10 (25 training epochs)	Image classification and recognition, specifically of airplanes (Sudharshan & Raj, 2018).	96.00%
Viola Jones	Haar-like features, AdaBoost, Cascade Classifier	Face detection and eye tracking (blinking detection, eye position) (Teja & Kumar, 2021)	Face: 98% Eyes: 94% With glasses: 82% Without glasses: 92%
PERCLOS ^a	EAR ^b computed from IR ^c images, blink monitoring	Determine driver drowsiness level through eyelid closure percentage (Teja & Kumar, 2021)	Not applicable (dependent on EAR)
YOLOv4 + CNN-29	Python, OpenCV, Darknet YOLOv4	Automatic detection of faults in solar panels (defective cells, hotspots, electrical failures) using thermal images captured by drone (Zou & G V, 2022)	Solar panel: 100% (confidence); Defective cells: 32% initially, up to 89% after retraining
YOLO	CNN (YOLO), bounding boxes and IOU ^d score	Object detection in the environment (e.g., bottles) to trigger alerts (Teja & Kumar, 2021).	98.00%
YOLOv4	Implemented with Darknet and optimized with TensorRT ^e on Jetson Nano platform	Real-time detection of Vietnamese vehicles (Minh, Mai, & Minh, 2021).	90.00%
		Real-time detection of Vietnamese license plates (Minh et al., 2021). Detection of objects and people (Liu, Liao, Chou, & Fan, 2021).	92.00% mAP ⁱ : 73.01%
MobileNetV2-SSD	Implemented with PyTorch and accelerated with TensorRT	Real-time detection of Vietnamese vehicles (Minh et al., 2021).	85.00%
		Real-time detection of Vietnamese license plates (Minh et al., 2021).	90.00%

^a PERcentage of eye CLOSure: metric used in drowsiness detection systems. ^b Eye Aspect Ratio: metric relating eye height and width to estimate openness. ^c Infrared: type of image captured via IR sensors. ^d Intersection over Union: metric measuring overlap between predicted and actual bounding boxes in object detection tasks. ^e TensorRT: NVIDIA inference optimization library that accelerates deep learning models for efficient execution on GPU (Graphics Processing Unit). ^f 2D: bidimensional, two-dimensional matrices in images. ^g MSAM (Modified Spatial Attention Module): spatial attention module that enhances the network's focus on relevant image regions. ^h PASCAL VOC 2007/2012: benchmark datasets for object detection (people, vehicles, animals). ⁱ mean Average Precision.

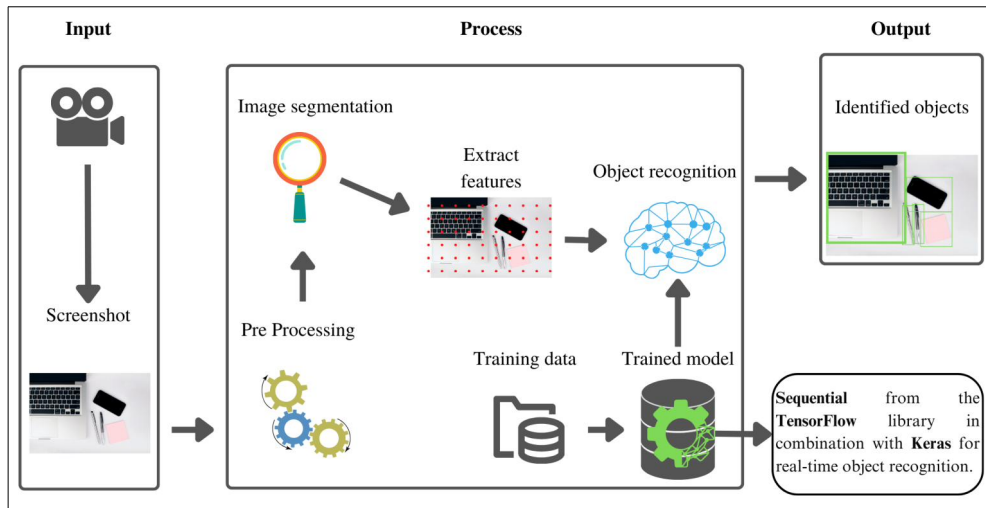


Fig. 2 Object recognition procedure used in the Torddis system.

assistance of the TDDT4IoTS tool (Guerrero-Ulloa, Carvajal-Suárez, Novais, Hornos, & Rodríguez-Domínguez, 2024), the IoT device of the system was designed.

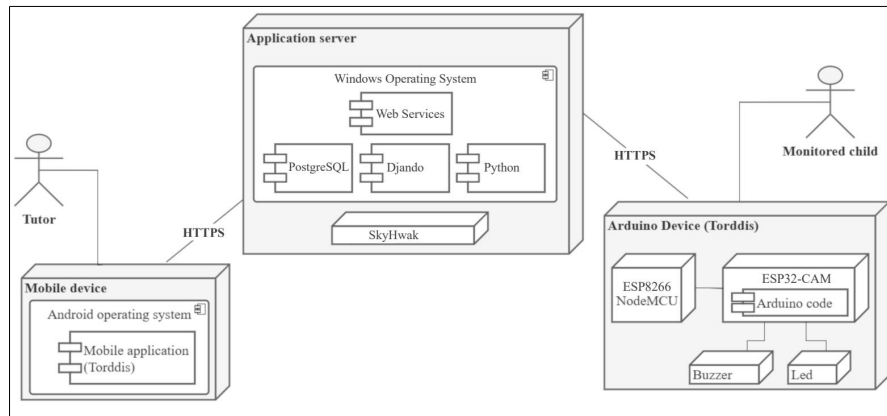


Fig. 3 Deployment diagram (nodes) of the Torddis system.

Figure 4 illustrates the structural design of the IoT device, highlighting the components used in its construction, which are:

1. ESP32-CAM board
2. ESP8266 NodeMCU board
3. Protoboard
4. Passive buzzer
5. Red LED

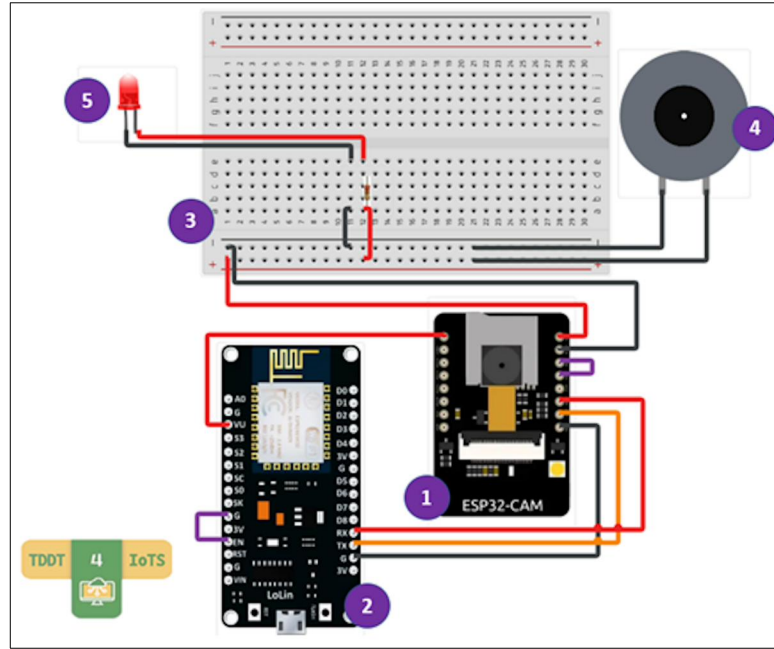


Fig. 4 Design of the IoT device of the Torddis system.

With the system architecture and coarse-grained functionalities already defined, the fine-grained requirements were elicited to ensure a successful implementation of Torddis. Figure 5 presents the use case diagram that clearly and specifically represents the problem domain.

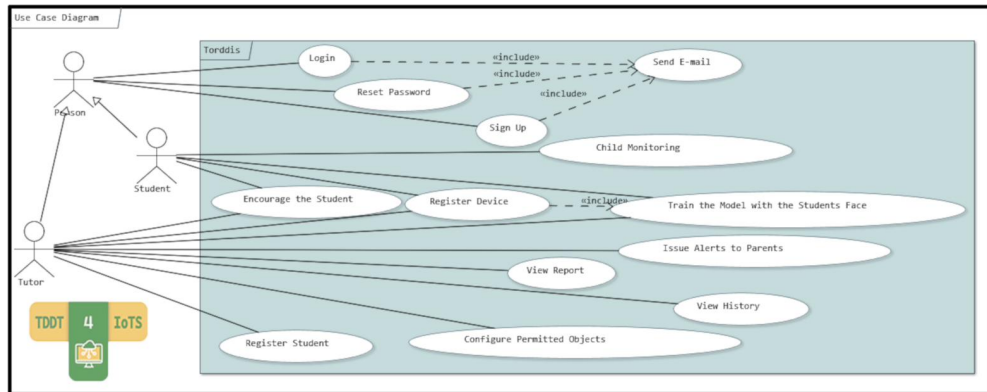


Fig. 5 Fine-grained use case diagram of the Torddis system.

Table 5 presents one of the fine-grained use cases in an extended format so that developers can implement the corresponding business logic and ensure the system meets the established specifications.

Table 5 Use Case: Register Torddis device

Use Case	Register Torddis device
Description	The tutor enters the MAC address of the Torddis device. The system registers the device in the database and displays a confirmation message.
Actors	Tutor
Preconditions	User is authenticated as a tutor.
Postconditions	The tutor's Torddis device is linked.
Main Flow	1. The use case begins when the tutor accesses the device-linking interface. 2. The system displays a form with the controls to link the device: MAC address input. 3. The tutor enters the MAC address and selects the save option (see Alt 1). 4. The system stores the data in the database (see E1). 5. The system provides feedback indicating a successful operation. 6. The use case ends when the system displays the facial training interface.
Alternative Flows	Alt 1. The tutor does not wish to register the Torddis device: 3. The use case ends when the tutor cancels the operation.
Exceptions	E1. Error when registering the Torddis device: 4. The system displays a message: <i>Error registering the Torddis device. Please try again.</i> 5. The tutor resumes from step 3 of the main flow.

Acknowledging technical limitations—such as object detection error rates and resource consumption—has made it possible to identify concrete areas for improvement. The implementation of lighter and optimized models, together with real-time calibration mechanisms, will ensure greater accuracy and performance of the system, even in less favorable environments (Falzetti & Cardone, 2024). Among the system modeling tools employed, structural models were used—such as class diagrams (see Figure 6)—which were generated directly by the TDDT4IoTS tool (Guerrero-Ulloa et al., 2024) based on the use cases that specify the system to be developed.

In addition, the AI computational models for monitoring learner events based on specified distraction parameters were obtained and refined by adjusting configurations such as the number of hidden layers and the number of training epochs. The unit and

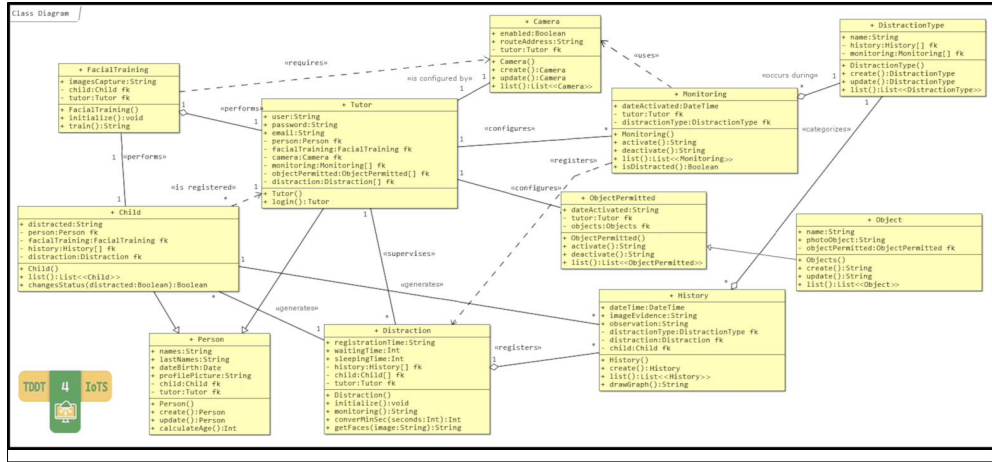


Fig. 6 Class diagram of the Torddis system.

integration tests that can be written in code (such as those related to database connections, web service testing, among others) were also automatically generated by the TDDT4IoTS tool (Guerrero-Ulloa et al., 2024). The data for unit testing (e.g., mobile application functions such as alert playback and drowsiness detection) were mainly obtained during the detailed requirements analysis phase, whereas the integration tests were derived during the system architecture design phase (i.e., communication between the ESP8286 NodeMCU and web services). Some of these integration tests were manually written and executed, such as the interface with the web services.

Figure 7 illustrates the relationship between the test data acquisition and the system life cycle phases. All system and acceptance tests were manually written and executed.

Tables 6, 7, 8, 9, and 10 summarize system test cases that were also used as acceptance tests, enabling verification that Torddis satisfies the defined requirements. These tests were authored and executed during the project—including validation with tutor users—to ensure alignment with the specifications.

Table 6 introduces the test case for tutor account creation, which serves as the system’s entry point. It is part of the documented system and acceptance test suite. The objective is to ensure that tutor registration is correctly recorded in accordance with the specifications (Sciarrà & Elfner, 2024).

Table 7 describes the test case that verifies signing in with a tutor account, explicitly included among the acceptance tests. It is also aligned with the non-functional requirement mandating encryption of personal and authentication data, thereby strengthening privacy during access.

Table 8 presents the test case for a student’s facial training; it requires a previously registered Torddis device and is included in the acceptance tests. The inclusion of this workflow is consistent with the observed performance of the recognition methods under controlled conditions.

Table 9 shows the test case that verifies linking the device to the tutor’s account via its MAC address; upon successful completion, the system displays the facial-training

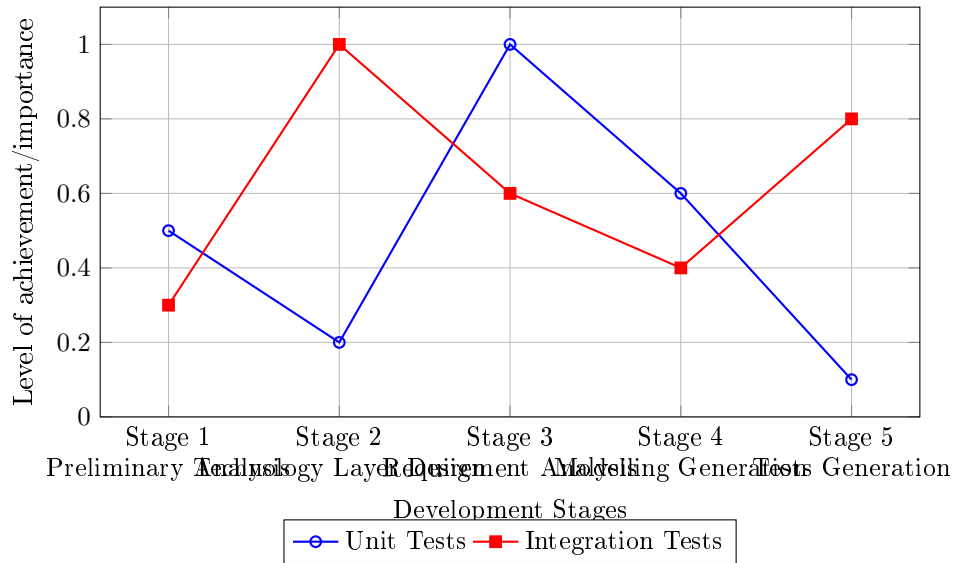


Fig. 7 Relationship between test data Acquisition and the Torddis life cycle phases.

Table 6 Test case for tutor account creation.

Test Case ID:	PU-1
Test Case Name:	Tutor Account Creation
Expected Outcome:	Successfully create a tutor account
Test Case Procedure	
Step No.	Step Description
1	Open the mobile application.
2	Select the “Sign Up” option.
3	Enter the registration data. • First name: Carlos Iván • Last name: Almeida Dueñas • Email: carlos.almeida2017@utec.edu.ec • Date of birth: 10/01/2000 • Username: calmeidad • Password: XXXXXX
4	Select the “Create Account” option.

interface. The steps confirm input of the MAC address and its persistence in the mobile application.

Table 10 documents the test case that ensures the tutor can enable or disable permitted objects for distraction monitoring, as specified by the functional requirement, and describes the effective procedure within the application.

Table 7 Test case for login.

Test Case ID:	PU-2
Test Case Name:	Login
Expected Outcome:	Successfully log in with a tutor account
Test Case Procedure	
Step No.	Step Description
1	Enter login credentials. • Username: calmeidad • Password: XXXXXX
2	Select the “Login” option.
3	The mobile application displays the tutor menu screen.

Table 8 Test case for facial training.

Test Case ID:	PU-3
Test Case Name:	Facial Training
Expected Outcome:	Successfully perform facial training for a student
Test Case Procedure	
Step No.	Step Description
1	Enter the “Child” module of the application.
2	Select the “Train” option for a child.
3	Register a device (see the test case shown in Table 9).
4	Select the “Train” option.
5	Position the child’s face in front of the device’s camera and wait for the facial training process to complete.

Table 9 Test case for device registration.

Test Case ID:	PU-4
Test Case Name:	Device Registration
Expected Outcome:	Successfully register the device in the mobile application
Test Case Procedure	
Step No.	Step Description
1	Select the “Register Device” option.
2	Enter the MAC address of the device.
3	Select the “Save” option.

Part of the software code was automatically generated by the TDDT4IoTS tool (Guerrero-Ulloa et al., 2024), and later refined and completed manually by the developers. The development team was responsible for delivering tested and functional software for each deliverable.

Table 10 Test case for configuring object usage.

Test Case ID:	PU-5
Test Case Name:	Object Usage Configuration
Expected Outcome:	Successfully configure object usage
Test Case Procedure	
Step No.	Step Description
1	Enter the “Objects” module of the application.
2	Search for the object to be activated/deactivated.
3	Select the object’s switch control to activate or deactivate it.

To complete the generation of the software code for the mobile application, the Android Studio IDE [Android Studio \(2024\)](#) was used with Java as the programming language, based on the code generated by the TDDT4IoTS tool. In addition, Python was used in conjunction with Visual Studio Code ([Microsoft Corporation, 2024](#)) to implement the web services, and C++ with the Arduino IDE ([Arduino, 2024](#)) to configure the hardware components of the IoT device within the Torddis system.

Consequently, this phase included both unit testing of the individual components of each deliverable and their respective integration tests. In summary, the outcome of this stage was a functional version of the software for each deliverable, thus validating its behavior according to the specified requirements. The various components of the mobile application, as well as the complete app, were deployed on a Xiaomi Note 10 Pro device running Android version 13. It is worth noting that the application’s architecture allows deployment on any mobile device running Android version 13 or higher, ensuring broad compatibility within the Android ecosystem.

Figure 8 displays screenshots of both the mobile application and the Torddis device, highlighting its intuitive design and the key features for real-time monitoring of student distractions. The IoT device, shown as well, integrates facial recognition and object detection technologies, providing comprehensive support to improve student concentration during schoolwork at home. Figure 8a shows the different options available to the tutor, while Figure 8b shows the list of students under a tutor’s supervision. On the other hand, Figure 8c presents the Torddis system device, which integrates the camera used to monitor the children while they perform their academic activities.

3 Maintenance Plan for the Torddis System

This supplementary section describes the preventive-maintenance plan devised for the Torddis prototype and the expected service life of its hardware components.

3.1 Maintenance

This phase has not yet been executed due to the limited timeframe required for the development of each deliverable, as well as the time allocated for the overall evaluation of the project. Nevertheless, the system must be maintained in accordance with the

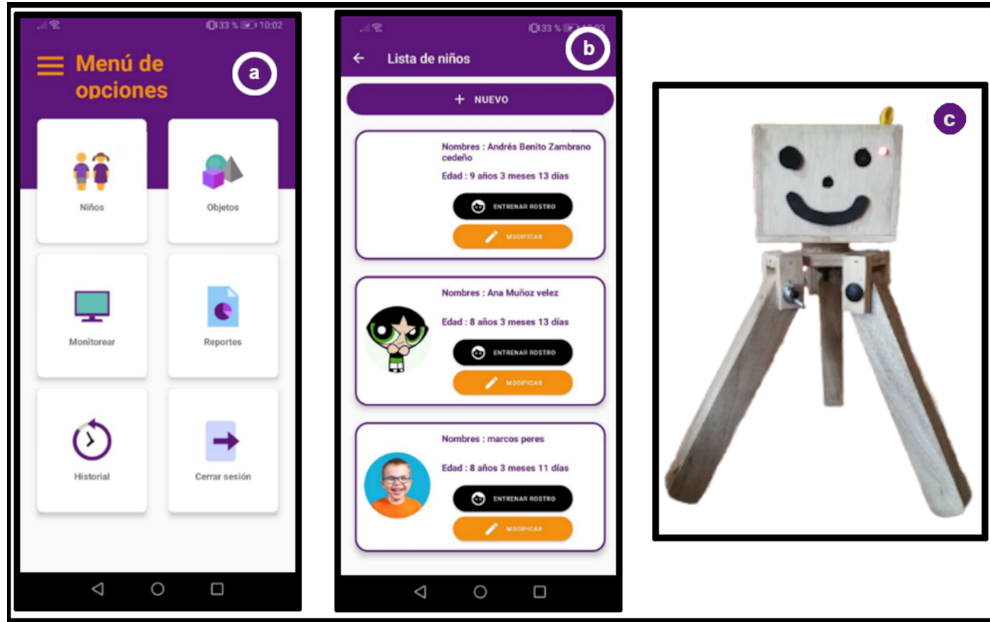


Fig. 8 Screenshots of the mobile application and the Torddis system device.

preventive guidelines and the expected lifespan of each of its components, in order to ensure proper functionality. Table 11 outlines some of the potential maintenance tasks. As indicated, each maintenance task—depending on its nature—should be carried out either on a quarterly or semiannual basis, although the actual frequency may vary depending on the operational environment in which the system is deployed.

Table 11 Definition of selected maintenance tasks.

No.	Maintenance Tasks	Quarterly	Semiannually
1	Internal cleaning of the device.	*	
2	Check whether the device components exhibit any imperfections or damage.		*
3	Inspect cable connections, condition of cables, connectors, etc.		*
4	Test the operation of the buzzer and LEDs.	*	
5	Test the functionality of the computing system (web application server, web services, mobile application, IoT device).		*

Preventive maintenance schedules are defined based on the specific component and its expected service life. Likewise, corrective maintenance for each physical component of the system should be performed before the end of its operational lifespan. The expected service life of each component is presented in Table 12. The lifespan of

resistors is expected to exceed that of the component with the highest durability in the system.

Table 12 Service life of the hardware components used in Torddis.

Component	Service Life (years)	Conditions That Reduce Service Life
ESP32-CAM	20 (ESPRESSIF, 2022)	Exposure to extreme temperatures, high humidity, dust, or mechanical vibrations.
ESP8266 Node MCU	20.83 (Amri & Setiawan, 2018) to 57 (Ardumotica, 2023 ; ESPRESSIF, 2018)	Extreme temperatures, high humidity levels, dust, voltage spikes, power supply noise, frequent power cycling, intensive WiFi usage at maximum power, firmware updates.
Passive Buzzer	3 (Huawha Electronics, 2023)	Excessive voltage, overcurrent, extreme temperatures, vibrations or mechanical shocks, moisture or corrosion, prolonged continuous operation.
LEDs	3 (Casamayor et al., 2015) to 14 (Green Lighting, 2024 ; Shenzhen Cary Technology Co., Ltd, 2024)	Component degradation, manufacturing conditions.
Resistors and others	∞ (expected)	High voltage stress (Simon et al., 2017).

The estimation of the components' service life has been primarily based on references from scientific literature. However, in the absence of relevant academic sources, the specifications provided by manufacturers and distributors have been considered, supplemented in certain cases with information obtained from specialized forums.

References

- Altameem, A., Kumar, A., Poonia, R.C., Kumar, S., Saudagar, A.K.J. (2021). Early identification and detection of driver drowsiness by hybrid machine learning. *IEEE Access*, 9, 162805-162819, <https://doi.org/10.1109/access.2021.3131601>
- Amri, Y., & Setiawan, M.A. (2018). Improving smart home concept with the internet of things concept using Raspberry Pi and NodeMCU. *Iop conference series: Materials science and engineering* (Vol. 325, pp. 1–10).
- Android Studio (2024). *Android mobile app development tools*. Retrieved from <https://developer.android.com/> (Accessed on June 28, 2024)
- Arduino (2024). *Software*. Retrieved from <https://www.arduino.cc/en/software> (Accessed on August 15, 2024)

- Ardumotica (2023). *Módulo WiFi NodeMCU V3 Lolin ESP8266 CH340*. Retrieved from <https://ardumotica.cl/producto/modulo-wifi-nodemcu-v3-lolin-esp8266-ch340/> (Accessed on December 06, 2024)
- Bikku, T., Viswanadha, L., Madhulatha, M., Jyothirgamai, P., Shabana, S., Sreedevi, S.S. (2022). Emotion detection and quotations using CNN. *2022 international conference on applied artificial intelligence and computing (icaaic)* (pp. 251–258). IEEE.
- Casamayor, J., Su, D., Sarshar, M. (2015). Extending the lifespan of LED-lighting products. *Architectural Engineering and Design Management*, 11(2), 105–122, <https://doi.org/10.1080/17452007.2013.834813>
- Cruz, J.P.N., Dimaala, M.L., Francisco, L.G.L., Franco, E.J.S., Bandala, A.A., Dadios, E.P. (2013, 3). Object recognition and detection by shape and color pattern recognition utilizing Artificial Neural Networks. *International conference of information and communication technology, icoict 2013* (pp. 140–144). IEEE.
- Darapaneni, N., Evoori, A.K., Vemuri, V.B., Arichandrapandian, T., Karthikeyan, G., Paduri, A.R., ... Madhavan, J. (2020). Automatic face detection and recognition for attendance maintenance. *2020 IEEE 15th international conference on industrial and information systems (iciis)* (pp. 236–241). IEEE.
- Deng, R., Jiang, M., Yu, X., Lu, Y., Liu, S. (2024). Does ChatGPT enhance student learning? a systematic review and meta-analysis of experimental studies. *Computers & Education*, 105224, <https://doi.org/10.1016/j.compedu.2024.105224>
- ESPRESSIF (2018, 3 13). *What is the lifetime expectancy of ESP-WROOM-32 module?* Retrieved from <https://www.esp32.com/viewtopic.php?t=4974#p21521> (Accessed on December 06, 2024)
- ESPRESSIF (2022, 1 28). *ESP32-powered DIY Wi-Fi security camera*. Retrieved from https://www.espressif.com/en/news/ESP32_Security_Camera (Accessed on December 06, 2024)
- Falzetti, P., & Cardone, M. (2024, 11). Promoting the improvement of education levels. *Icieri2024 proceedings* (pp. 8416–8423). IATED.
- Garcia, R.A.C., Lacayanga, R.P., Cruz, F.R.G. (2021). Application of Artificial Intelligence in adaptive face recognition system. *2021 IEEE 11th international conference on system engineering and technology (icset)* (pp. 263–268). IEEE.

- Girmay, S., Samsom, F., Khattak, A.M. (2021). AI based login system using facial recognition. *2021 5th cyber security in networking conference (csnet)* (pp. 107–109). IEEE.
- Goel, P., & Agarwal, S. (2012). Hybrid approach of Haar Cascade Classifiers and geometrical properties of facial features applied to illumination invariant gender classification system. *2012 international conference on computing sciences* (pp. 132–136). IEEE.
- Green Lighting (2024). *The lifespan of an LED bulb*. Retrieved from <https://greenlighting.co.uk/the-lifespan-of-an-led-bulb/> (Accessed on December 06, 2024)
- Guerrero-Ulloa, G., Carvajal-Suárez, D., Novais, P., Hornos, M.J., Rodríguez-Domínguez, C. (2024). Test-driven development tool for IoT systems. *IEEE Software*, 1–9, <https://doi.org/10.1109/ms.2024.3479880> (PrePrint)
- Huawha Electronics (2023, 3 28). *What is the lifespan of smd piezo buzzer?* Retrieved from https://www.jshnbuzzer.com/new_detail/nid/85649.html (Accessed on December 06, 2024)
- I, S.S., Ramli, R., Azri, M.A., Aliff, M., Mohammad, Z. (2022, 5). Raspberry Pi based driver drowsiness detection system using Convolutional Neural Network (CNN). *Ieee 18th international colloquium on signal processing & applications, cspa 2022* (pp. 30–34). IEEE.
- Ilić, V., Batić, Đ., Mirković, M., Vukmirović, S., Ćulibrk, D., Bosakov, G., Popović, I. (2021). Automatic emotion detection as a teaching aid in online knowledge assessment. *2021 20th international symposium infoteh-jahorina (infoteh)* (pp. 1–5). IEEE.
- Jaiswal, A., Krishnama Raju, A., Deb, S. (2020). Facial emotion detection using Deep Learning. *2020 international conference for emerging technology (incet)* (pp. 1–5). IEEE.
- Kedari, P., Kapile, M., Kadole, D., Jaikar, S. (2021). Face emotion detection using Deep Learning. *2021 2nd international conference on advances in computing, communication, embedded and secure systems (access)* (pp. 118–123). IEEE.
- Kumar, M., & Srivastava, S. (2021). Emotion detection through facial expression using Deep Learning. *2021 5th international conference on information systems and computer networks (iscon)* (pp. 1–4). IEEE.
- Lalitha, S.K., Aishwarya, J., Shivakumar, N., Srilekha, T., Kartheek, G.C.R. (2021). A deep learning model for face expression detection. *2021 international conference on recent trends on electronics, information, communication & technology*

- (*rteict*) (p. 647-650). IEEE.
- Le, C., & Mohd, T.K. (2022). Facial detection in low light environments using OpenCV. *2022 ieee world ai iot congress (aiiot)* (pp. 624–628). IEEE.
- Liu, F.-Y., Liao, C.-L., Chou, P.-W., Fan, Y.-C. (2021). Objects detection deep learning system based on 2-D winograd Convolutional Neural Network. *2021 ieee 10th global conference on consumer electronics (gcce)* (pp. 454–455). IEEE.
- Microsoft Corporation (2024). *Visual studio code - code editing. redefined*. Retrieved from <https://code.visualstudio.com/>
- Minh, H.T., Mai, L., Minh, T.V. (2021). Performance evaluation of deep learning models on embedded platform for Edge AI-based real time traffic tracking and detecting applications. *2021 15th international conference on advanced computing and applications (acompa)* (pp. 128–135). IEEE.
- Sciarra, A., & Elfner, H. (2024). SMASH as an event generator for heavy-ion collisions. *Frontiers in Physics*, 12, 1502621, <https://doi.org/10.3389/fphy.2024.1502621>
- Shanmugam, K.S., Badruddin, N., Asirvadam, V.S. (2022). Comparative study of state-of-the-art face landmark detectors for eye state classification in subjects with face occlusion. *2022 ieee 4th eurasia conference on biomedical engineering, healthcare and sustainability (ecbios)* (pp. 24–27). IEEE.
- Shenzhen Cary Technology Co., Ltd (2024). *LED lifespan: light emitting diode lifetime & MTBF - knowledge*. Retrieved from <https://www.carylighing.com/info/led-lifespan-light-emitting-diode-lifetime-53797730.html> (Accessed on December 06, 2024)
- Silva, P., Sudasinghe, S.A.V.D., Hansika, P.D.U., Gamage, M.P., Gamage, M.P.A.W. (2021). AI-Based e-learning solution to motivate and assist primary school students. *2021 3rd international conference on advancements in computing (icac)* (pp. 294–299). IEEE.
- Simon, C., Rebillat, F., Camus, G. (2017). Electrical resistivity monitoring of a SiC/[Si-B-C] composite under oxidizing environments. *Acta Materialia*, 132, 586–597, <https://doi.org/10.1016/j.actamat.2017.04.070>
- Singh, S., & Nasoz, F. (2020). Facial expression recognition with Convolutional Neural Networks. *2020 10th annual computing and communication workshop and conference (ccwc)* (pp. 0324–0328). IEEE.
- Sudharshan, D.P., & Raj, S. (2018). Object recognition in images using convolutional neural network. *2018 2nd international conference on inventive systems and*

- control (icisc)* (pp. 718–722). IEEE.
- Teja, K.B.R., & Kumar, T. (2021). Real-time smart drivers drowsiness detection using DNN. *2021 5th international conference on trends in electronics and informatics (icoei)* (pp. 1026–1030). IEEE.
- Zhang, J., & Yu, H. (2022, 9). Improving the facial expression recognition and its interpretability via generating expression pattern-map. *Pattern Recognition*, 129, 108737, <https://doi.org/10.1016/j.patcog.2022.108737>
- Zou, J.-T., & G V, R. (2022). Drone-based solar panel inspection with 5G and AI technologies. *2022 8th international conference on applied system innovation (icasi)* (pp. 174–178).